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Find the general solution of the given equation.

$$\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 100y = 0$$

$y = e^{rx}$ is a solution to the second-order homogeneous differential equation of the form $ay'' + by' + cy = 0$ if and only if r is a solution to $ar^2 + br + c = 0$. The equation $ar^2 + br + c = 0$ is called the auxiliary equation to $ay'' + by' + cy = 0$.

Write the auxiliary equation for the given differential equation.

$$r^2 - 20r + 100 = 0$$

Find the roots of the auxiliary equation to determine the form of the general solution to $\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 100y = 0$.

$$r^2 - 20r + 100 = 0$$

$$(r - 10)(r - 10) = 0$$

Factor.

$$r = 10$$

Solve.

The auxiliary equation has one real (repeated) root, $r = 10$. Therefore, the general solution to the given differential equation is of the form shown below.

$$y = c_1 e^{rx} + c_2 x e^{rx}$$

Write the general solution to $\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 100y = 0$.

$$y(x) = c_1 e^{10x} + c_2 x e^{10x}$$